

Govind Mudavadkar

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EDUCATION

Northeastern University, Boston, Massachusetts (GPA: 3.8/4.0)

Sep 2023 - Dec 2025

Master's of Science in Computer Engineering

University of Pune, India (GPA: 3.8/4.0)

Jun 2019 - May 2023

Bachelor of Engineering, Computer Science and Engineering

EXPERIENCE

AI/ML Engineer

Jan 2026 – Present

Pfizer Inc

Groton, CT

- Architected **agentic AI platform** for enterprise data ingestion, real-time question answering, on **AWS**(S3, Bedrock, EC2) saving **800+** hours annually
- Engineered modular multimodal AI pipelines for **text and image** retrieval on AWS to support faster real-time inference, reducing latency from **2 minutes to 50 seconds** through retrieval tuning and model orchestration and pipeline optimization
- Designed **LangGraph**(Langchain ecosystem) based conversational **agent workflows** with reusable nodes for retrieval, multimodal input handling, and tool including automated document generation and prompting via **MCP** (Model Context Protocol); reduced manual work overhead annually by **25%**
- Built a **AI assistant platform** using Python, FastAPI, React(TypeScript), and **SQL**, deployed on **AWS** to deliver real-time inference to end users, reduced latency by **25%** through A/B testing and system evaluation
- Containerized and orchestrated platform using **Docker and Kubernetes** for scalability, serving **50+** scientists with improved inference latency by **25%**
- Served as **AI/ML SME** to **10+** scientists, translating business needs into scalable GenAI infrastructure, slashing time to production by **33%** through modular design that standardized use cases

AI/ML Automation Engineer Co-Op

Jan 2025 – Sep 2025

Pfizer Inc

Groton, CT

- Architected an in-house multimodal AI platform combining **LLMs**, **RAG**, **vector DBs**, and knowledge graphs, supporting **400+** daily queries across automated document intelligence and data exploration workflows
- Built production **RAG pipelines** using embeddings, vector search, and metadata filtering to improve enterprise document search, increasing retrieval relevance by **30%**
- Enhanced LLM output quality by **27%** via applying custom chunking, multi-step prompt chaining, and retrieval optimization, reducing hallucinations in production AI workflows
- Deployed transformer based (MolNexTR) **NLP pipeline** (HuggingFace) to parse and classify domain-specific structured inputs, cross-referenced against knowledge graph for semantic matching and retrieval improving information retrieval accuracy by **27%**
- Developed **bayesian optimization** workflow on HPC infrastructure to accelerate search, parameter tuning, improving optimization efficiency by **90%**
- Implemented a **computer vision** system with YOLO, Faster R-CNN, AWS S3 on Dataiku DSS for automatic video labeling and workflow automation
- Co-authored two research abstracts in **AICHe 2025**, contributing to productionize AI workflows and scalable data warehousing

Data Scientist Research Assistant

Feb 2024 – Dec 2024

Northeastern University

Boston, MA

- Built **PyTorch** based deep learning (**DL**) models for image classification and diagnostic prediction, improving accuracy by **18%**
- Exported **ML models** to ONNX, ran INT8 inference to batch-process **500K+** images, improving throughput by **2.5X** without diagnostic accuracy loss
- Accelerated models(ViTNet, ResNet, Ensemble) training workflows using distributed **GPU workloads**(Bash, Shell Script) on Oracle Cloud computing platform, reducing training time from **8 Hrs to 4 Hrs**
- Engineered end-to-end image processing pipeline using machine learning (**ML**) models,AWS(S3, SageMaker) to automate storage and inference for large image datasets, increasing throughput by **2X**

Software Development Engineer

Jan 2022 - Jul 2022

BMC Software Inc

Pune, India

- Developed a custom VS Code extension using **Vue.js**, **WebView API** and **Spring Boot** backend to automate **microservices** metadata retrieval, reducing manual data entry by **63%** and eliminating context switching between IDEs and databases
- Developed Java based REST APIs with PostgreSQL to speed up service discovery from minutes to seconds and improve developer workflow efficiency
- Collaborated in Agile teams across backend and developer tooling, partnering with engineers and stakeholders to ship productivity improvements

SKILLS

Programming & Scripting: Python, Java, R, C++, SQL, Scala, Bash, linux

Machine Learning & AI: Numpy, Pandas, Natural Language Processing, Ensemble, HuggingFace, Transformers, PyTorch, Tensorflow, AI Agents (LangGraph), Langchain, Langfuse LLMs, Sequence Models, Vision-Language Models (VLMs), MCP, Computer Vision, Statistics

Databases: PostgreSQL, MySQL, NoSQL, MongoDB, Spark, Vector DBs(Pinecone, Milvus, ChromaDB)

Web-Development: TypeScript, REST APIs, Node.js, Agile, FastAPI

Cloud & DevOps: AWS (S3, SageMaker, Lambda, EC2), GitHub, CI/CD, Docker, Kubernetes, Git, Azure(AI Services, AI Foundry), (K8s manifests)

PUBLICATIONS

- **KnovaAI: Advanced Multimodal AI for Chemistry- Aware Document Intelligence** | *AICHe 2026*
- **Machine Learning-Driven API Development via Automated Platforms** | *AICHe 2025* | [Link](#)
- **Gastric Cancer Detection with Ensemble Deep Learning** | *MDPI Diagnostics Journal 2024* | [Link](#)
- **Additional 4+ peer-reviewed research available via Google Scholar link** | *Google Scholar* | [Link](#)